

Sahana Aster KŌWHAI, Ph.D.

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Associate Professor of Palaeoclimatology, School of Earth and Ice Studies
Kārearea University,
Southern Ocean Research Centre, Te Waka Wānanga, Ōtepoti Dunedin, Aotearoa New Zealand

SHORT BIO

Sahana Aster Kōwhai is a **fictional** palaeoclimatologist whose *research* reconstructs Southern Ocean change from marine sediment archives.

RESEARCH FOCUS

Southern Ocean palaeoceanography across glacial terminations, with an emphasis on multiproxy reconstructions, reproducible chronology and equitable Antarctic field partnerships.

ACADEMIC APPOINTMENTS / EXPERIENCE

- Associate Professor of Palaeoclimatology**
Kārearea University, School of Earth and Ice Studies
Feb 2021 – Present
Ōtepoti Dunedin, Aotearoa New Zealand
 - Cross-appointed affiliate of the Southern Ocean Research Centre, Te Waka Wānanga.
- Senior Lecturer in Marine Geology**
University of Southern Tasmania, Institute for Marine and Southern Ocean Studies
Jul 2016 – Jan 2021
nipaluna Hobart, Australia

EDUCATION

- Ph.D. in Marine Geology**
Commonwealth University of Australia
2010 – 2014
Canberra, Australia
 - Thesis: Foraminiferal Mg/Ca and Southern Ocean warming across Termination I.
- M.Sc. in Earth Sciences (First Class Honours)**
Cape Maritime University
2008 – 2009
Cape Town, South Africa
- B.Sc. in Geology and Oceanography**
Madras Institute of Ocean Sciences
2005 – 2008
Chennai, India

TEACHING

- Course coordinator and lecturer**
Kārearea University
2021 – Present
Ōtepoti Dunedin, Aotearoa New Zealand

Course	Programme	Topics	Points
GEOL 273	B.Sc. Geology	Palaeoclimate archives; proxy calibration	18
GEOL 472	B.Sc. (Hons)	Marine cores; age-depth modelling	18
OCEN 405	M.Sc. Oceanography	Southern Ocean circulation; uncertainty	20

EDITORIAL, REVIEWING & CONFERENCE SERVICE

- Associate Editor**
Palaeoceanography and Climate Systems, Marine palaeoclimate
2023 – Present
- Associate Editor**
Example Journal of Palaeoclimate Studies, 2022–2024 [IF: 3.4, Q1]
- Scientific Steering Committee**
Southern Ocean Climate Observatory Consortium, 2022–2026
- Panel reviewer**
Aotearoa Research Council — Te Kāhui Pūtaiao
2021 – Present
- Reviewer**
Quaternary Palaeoenvironment Reviews

RESEARCH PROJECTS

- TIDE — Termination Ice Dynamics and Ecosystems

Aotearoa Research Council — Te Kāhui Pūtaiao, Principal investigator

Multiproxy reconstruction of Southern Ocean fronts across Termination II, with voyage support from the Southern Archive Voyage Consortium.

2024 – 2027
- Open Core Chronologies

Austral Polar Science Partnership, Co-investigator

Reusable age-depth models and provenance for legacy sediment cores.

2022 – 2025

AWARDS & SCHOLARSHIPS

- Early Career Research Excellence Award

Geoscientific Society of Aotearoa

2019
- ICPR Visiting Fellowship

International Council for Polar Research

2017

Cambridge, United Kingdom

STUDENT SUPERVISION (THESES & INTERNSHIPS)

Current supervision: projects span marine geochemistry, chronology and open polar data.

Level	#	Notes (scope and international experience)
Doctoral candidates	4	Primary supervisor for two; co-supervisor for two
M.Sc. and honours researchers	9	Completed and current research dissertations

ADDITIONAL INFORMATION

Languages: Tamil (Native), English (Fluent), te reo Māori (Intermediate learner)

PUBLIC ENGAGEMENT

- Sediment stories: reading climate from ocean cores

Kōwhai Coast Learning Collective, Community workshop for secondary-school science teachers

Sep 2025

Ōamaru, Aotearoa New Zealand
- Core archive open day

Aotearoa Marine Memory Trust, Public demonstration of core scanning and age-model uncertainty

Mar 2024

Porirua, Aotearoa New Zealand
- Antarctic archives teacher residency

Polar Evidence Studio, Short course on proxy evidence for secondary-school teachers

Jul 2023

Ōtepoti Dunedin, Aotearoa New Zealand

TALKS & PRESENTATIONS

I=INVITED, C=CONFERENCE

Invited and keynote talks

- [I1] S. A. Kōwhai. “What an open sediment record should remember”. World Congress on Palaeoceanography (Bergen, Norway). Invited keynote. June 18, 2026.

Conference presentations

- [C1] S. A. Kōwhai and M. Te Rangi. “First results from the Pukaki Rise coring transect”. Aotearoa Antarctic Science Symposium (Te Whanganui-a-Tara Wellington, Aotearoa New Zealand). Oral presentation. Aug. 21, 2025.
- [C2] S. A. Kōwhai. “Age-depth ensembles without hidden decisions”. Australasian Quaternary Research Society Biennial Meeting (nipaluna Hobart, Australia). Poster presentation. Dec. 4, 2024. URL: <https://talks.example.edu/kowhai2024ensembles>.
- [C3] S. A. Kōwhai. “From core description to an auditable climate record”. Southern Archives Methods Forum (Ōtepoti Dunedin, Aotearoa New Zealand). Sept. 14, 2023.

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, B=BOOKS, D=DATA, R=UNDER REVIEW

Journal articles (peer-reviewed)

- [J1] S. A. Kōwhai, M. Te Rangi, and C. Okafor. “Southern Ocean front migration recorded across two glacial terminations”. In: *Palaeo-Environments, Palaeoclimates, Palaeoecology* 612 (2026), p. 111001.
- [J2] S. A. Kōwhai, H. Williams, and T. Ndlovu. “Transparent age-depth ensembles for Southern Ocean sediment archives”. In: *Earth’s Past Climate* 20.8 (2024), pp. 1701–1724. URL: <https://publications.example.edu/kowhai2024chronology>.

Conference papers (peer-reviewed)

- [C1] S. A. Kōwhai and M. Te Rangi. “Automated tephra isochron matching in Southern Ocean sediment cores”. In: *Proceedings of the International Palaeoceanography Symposium*. Bergen, Norway, 2023, pp. 145–152. DOI: [10.5555/palaeo.2023.0145](https://doi.org/10.5555/palaeo.2023.0145).

Books and chapters

- [B1] S. A. Kōwhai. "Proxy agreement and disagreement in high-latitude marine archives". In: *Southern Ocean Palaeoceanography: Methods and Records*. Ed. by L. Navarro and K. Adeyemi. Cape Town: Aster Academic Press, 2023, pp. 81–108.
- [B2] S. A. Kōwhai and C. Okafor. "Southern Ocean Foraminiferal Assemblages as Thermal Proxies". In: *Advances in Marine Palaeoecology*. Ed. by L. Navarro. Aster Academic Press, 2022. Chap. 4, pp. 89–112.
- [B3] S. A. Kōwhai and M. Te Rangi. *Field Notes for Reciprocal Polar Science*. Ōtepoti Dunedin: Koru Open Books, 2021.

Data archives

- [D1] S. A. Kōwhai and C. Okafor. *Pukaki Rise multiproxy core chronology, version 1.0*. Aotearoa Polar Data Commons, 2025. URL: <https://data.example.edu/pukaki-rise-v1>.

Manuscripts under review

- [R1] S. A. Kōwhai and M. Te Rangi. "Shelf-water change at the Pacific Antarctic margin". In: *Quaternary Palaeoenvironment Reviews* (2028).
- [R2] S. A. Kōwhai and T. Ndlovu. "Sea-ice retreat and productivity at the East Antarctic margin". In: *The Polar Cryosphere* (2027).

Example record. This curriculum vitae describes a fictional scholar and is supplied only to demonstrate the template.